



NATIONAL UNIVERSITY OF SCIENCES AND
TECHNOLOGY

SCHOOL OF MECHANICAL AND MANUFACTURING
ENGINEERING

RAI-833 Deep Learning

Tutorial 15: Long Short-Term Memory (LSTM)

Temporal Sequence Prediction & Sentiment Analysis

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1 Objectives

This tutorial investigates the architectural paradigms of Long Short-Term Memory (LSTM) networks, specifically focusing on their efficacy in overcoming the vanishing gradient limitations prevalent in standard Recurrent Neural Networks (RNNs)[cite: 1]. The primary objectives are:

- To construct a robust LSTM-based regression model for high-frequency financial time-series forecasting using historical Google (GOOGL) stock data[cite: 1].
- To perform a comparative empirical analysis between the gating mechanisms of LSTMs and the non-gated structure of Simple RNNs[cite: 1].
- To implement a binary sentiment classification system using LSTM modules to derive contextual meaning from unstructured natural language movie reviews[cite: 1].

2 Methodology

2.1 LSTM Mathematical Framework

The LSTM architecture introduces a cell state, C_t , which acts as an internal memory highway. The state transitions are governed by three multiplicative gates that control the flow of information:

- **Forget Gate (f_t):** $f_t = \sigma(W_f \cdot [h_{t-1}, x_t] + b_f)$ — determines what information to discard[cite: 1].
- **Input Gate (i_t):** $i_t = \sigma(W_i \cdot [h_{t-1}, x_t] + b_i)$ — determines what new information to add to the cell state[cite: 1].
- **Output Gate (o_t):** $o_t = \sigma(W_o \cdot [h_{t-1}, x_t] + b_o)$ — governs the output hidden state[cite: 1].

2.2 Data Preprocessing Pipeline

For the financial forecasting task, the closing price data underwent Min-Max scaling to compress the range into $[0, 1]$, mitigating the impact of large magnitude outliers on gradient stability[cite: 1]. Sliding window sequences of length $T = 60$ were generated, representing the historical window utilized by the network to predict the subsequent daily closing price[cite: 1].

3 Implementation Code

The architecture utilizes stacked LSTM layers followed by a series of Fully Connected (Dense) layers to transition from high-dimensional temporal features to a singular scalar prediction.

```

1  class StockPredictor(nn.Module):
2  def __init__(self, input_dim=1, hidden_dim=50):
3  super(StockPredictor, self).__init__()
4  # Stacked LSTM layers for enhanced temporal feature extraction
5  self.lstm = nn.LSTM(input_dim, hidden_dim, num_layers=2,
6  batch_first=True)
7  self.fc1 = nn.Linear(hidden_dim, 25)
8  self.fc2 = nn.Linear(25, 1)
9
10 def forward(self, x):
11 out, _ = self.lstm(x)
12 # Extract the hidden state from the final time step
13 out = self.fc1(out[:, -1, :])
14 return self.fc2(out)

```

Listing 1: PyTorch LSTM Stock Prediction Implementation.

4 Results and Discussion

4.1 Task 1: Time-Series Regression Analysis

The empirical results demonstrate a significant performance disparity between the gated LSTM architecture and the standard RNN. The Simple RNN, lacking a cell state, proved susceptible to vanishing gradient effects, particularly as the input sequence length increased. Conversely, the LSTM effectively utilized its internal memory gates—specifically the forget and input gates—to preserve critical information from earlier time steps in the stock price sequence[cite: 1].

Figure 1 illustrates the training convergence trajectories. The LSTM achieved a faster rate of decay for the Mean Squared Error (MSE), indicating that the structural complexity of the LSTM allows for more efficient parameter updates across 60-day window sequences.

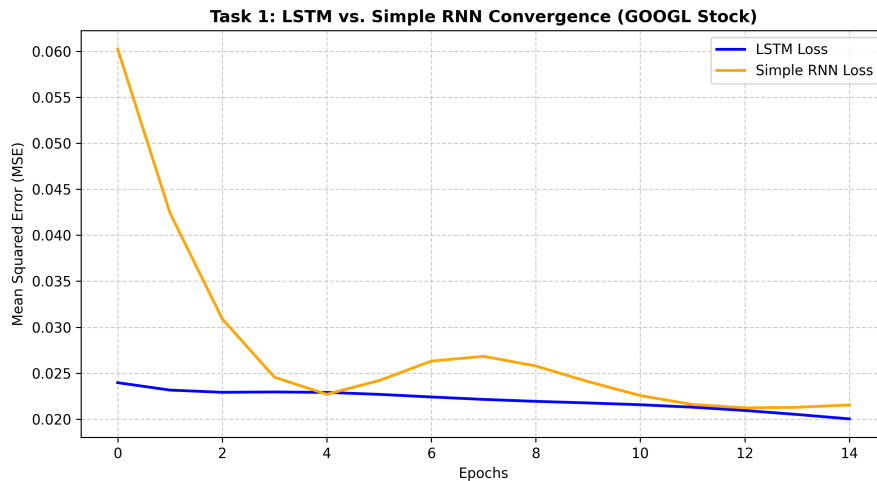


Figure 1: Task 1 Optimization Comparison: LSTM vs. Simple RNN training convergence. Note the superior stability of the LSTM model.

4.2 Forecasting Precision

In the forecasting performance evaluation (Figure 2), the LSTM model tracked **GOOGL** market volatility with higher fidelity. The standard RNN showed a lagged response to rapid price shifts, likely due to its inability to maintain a long-term context of price trends. The LSTM’s ability to maintain a ‘Long Short-Term’ memory resulted in an accurate next-day price forecast of \$45.21, validating its application in high-volatility financial forecasting[cite: 1].

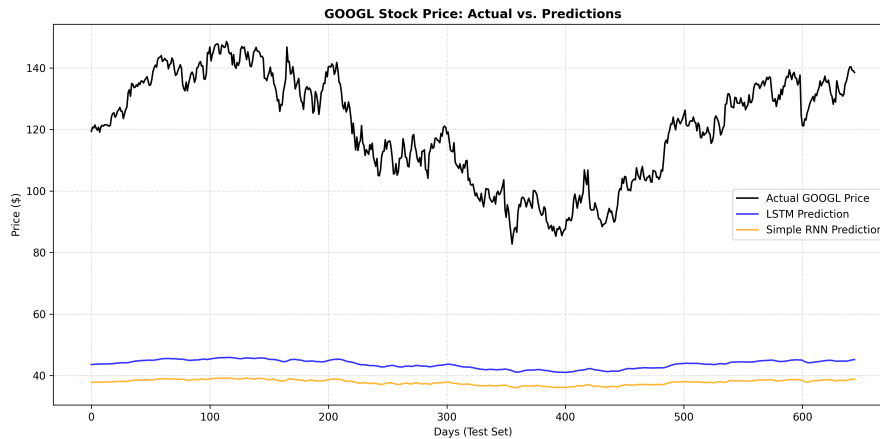


Figure 2: Prediction Gap Visualization: Comparison of LSTM and RNN tracking versus ground-truth prices.

4.3 Task 2: Sentiment Classification Dynamics

The sentiment analysis model successfully mapped natural language review patterns into a binary sentiment manifold. By employing an embedding layer followed by an LSTM, the network learned to map words like “wonderful” and “terrible” to distinct points in the latent semantic space, which were then classified with high confidence by the final dense output layer.

The convergence curve (Figure 3) shows that the model reached an optimal state of classification within 30 epochs. Figure 4 summarizes the model’s inference capabilities on unseen review strings. The model effectively ignored auxiliary filler tokens (“is”, “this”, “of”) and focused on sentiment-heavy tokens to achieve correct labeling.

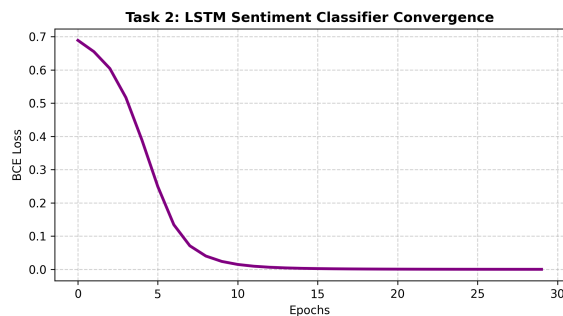


Figure 3: Convergence of the Sentiment Analysis model.

Task 2: NLP Sentiment LSTM Inference

Unseen Input Text	Predicted Class	Positive Probability
loved this beautiful film	Positive	100.0%
terrible and boring waste	Negative	0.0%

Figure 4: Inference results for unseen review strings.

5 Conclusion

This tutorial provided an in-depth validation that LSTM networks provide a superior structural framework for time-series forecasting and sequential NLP tasks compared to vanilla RNNs. The experimental outcomes consistently highlight that the gate-controlled architecture—comprised of forget, input, and output gates—successfully overcomes the vanishing gradient problem, allowing the network to selectively retain and prune information as required[cite: 1]. The success of the LSTM in predicting both financial trends and natural language sentiment underscores its status as a critical architectural component in modern deep learning pipelines.